The Three-Center Electron-Repulsion Integral Is an Elliptic Bessel Moment: A Genus-One Radial Core at the Third Center in Momentum-Space Slater Theory*

J. Loutey[1]

[1] *Independent Researcher, Kent, Washington*

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Two-center electron-repulsion integrals over Slater-type orbitals close in elementary functions plus the exponential integral, the logarithm, and Euler's γ—a genus-zero transcendence class—which is why the two-center problem has been solved since Roothaan and Ruedenberg. The genuine three-center integral has resisted a closed form for seventy years, and the standard verdict is that it is simply hard: reduce it to an infinite series or a one-dimensional quadrature and be done. We give it a structural reading instead. Evaluating the two-electron three-center integral $(XY|XZ)$ over $1s$ orbitals in momentum space—the Fock projection under which Coulomb–Sturmians become hyperspherical harmonics, the setting native to the Averys' generalized-Sturmian program—the three centers enter as three *phases* rather than three foci, so the coordinate obstruction (prolate spheroidal coordinates admit exactly two foci) dissolves. **[MEASURED]** The resulting momentum integral reproduces the exact integral (agreement 1.9×10^{-14} with McMurchie–Davidson on the same Gaussian fits; with the true Slater density, 4.3×10^{-7} against the fitted reference, which itself sits 7×10^{-8} above the Slater value), and its angular part collapses to a single spherical Bessel kernel. **[MEASURED / classical]** What remains—the radial content—is a two-scale Bessel moment whose period is a *complete elliptic integral* (verified to 31 digits), living on the genus-one curve $y^2 = (c_1k^2 + 1)(c_2k^2 + 1)$; it degenerates to the genus-zero two-center case exactly on the diagonal $c_1 = c_2$. **[SYMBOLIC + MEASURED]** The one-mass slice is the Laplace transform of the elliptic differential—an elliptic deformation of the Bessel K_0, reducing to $K_0(D)$ when one scale decouples ($\rho \to 0$) and to the complete elliptic integral K at zero separation—and satisfies an explicit fourth-order Picard–Fuchs equation with polynomial coefficients, whose characteristic exponents are the four branch points of the curve. **[OBSERVATION]** This places a molecular integral inside the modern theory of Bessel moments and two-mass elliptic Feynman integrals; a targeted literature search finds that connection unmade in either field. **[OPEN]** We do *not* give the finite closed form, and we show the natural route to it is obstructed: the elliptic-dilogarithm (unequal-mass sunrise) mechanism does not engage, because the modulus Picard–Fuchs source is not an elementary subtopology but lies in the integral's own Gauss–Manin D-module—the branch-point boundary that would supply the sunrise's tadpole vanishes identically. **[OBSERVATION]** The object is therefore an apparently unclaimed elliptic transcendent one level *above* the elliptic-dilogarithm tower, a period of a rank-four irregular connection. **[SYMBOLIC + MEASURED]** That connection is *irreducible*—it is the Fourier–Laplace transform of an irreducible connection, and its monodromy admits no invariant coordinate subspace—so no closed form arises through factorization; **[OPEN]** whether the irreducible object admits a transcendental closed form we leave open. The message is not a solution but a relocation: the third center does not make the radial core messier within the two-center class, it raises the radial core's transcendence one genus, past the reach of the sunrise's elliptic-dilogarithm mechanism. **[MEASURED]** The integrated three-center value leaves the two-center class too, as a function of the bond angle: its bond-angle operator, built from exact coefficients modulo primes, keeps a piece of order six with no first-order factor at two geometries, where a two-center-class form would factor completely (assuming its constants independent; the class of the value at one geometry is not decided). The rungs beyond the third center—a fourth center, the three-electron chain's K3 surface, four linked densities and the first period beyond elliptic ones, and the ring of three—are taken up in Paper 63 [1], which lifts the ladder from $1s$ orbitals to every exponential-type orbital.

I. THE TWO-CENTER CLASS, AND THE WALL BEYOND IT

The electron-repulsion integral over Slater-type orbitals on two centers is one of the oldest solved problems in molecular quantum mechanics [32]. In prolate spheroidal coordinates, with the two nuclei at the foci, the Neumann expansion of $1/r_{12}$ terminates against the orbital angular momenta, and every integral reduces to elementary functions together with the exponential integral E_1, the logarithm, and Euler's constant γ. In the transcendence bookkeeping of the Geometric Vacuum program (Paper 18 [50]), these are *embedding-tier* constants—the cost of the continuum two-center geometry rather than content of the discrete graph—and, decisively, they sit at transcendence *weight one*, and no dilogarithm appears, even though the ordered radial integral has the simplex shape that generically produces one. In the direct-Coulomb $(AA|BB)$ class—and there

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alone—even the prefactor π cancels against the Coulomb 4π, leaving a rational combination of $\{1,\, e^{-\lambda R}\}$ whose zeros Lindemann–Weierstrass separates; that is what lets *that block* be *decided* rather than merely counted (Paper 58 [52]). It does not extend to the three genuinely cross classes, which carry E_1, ln and γ seeds and remain counted—weight one is not the same as decidable.

The genuine three-center integral has no such story. Prolate spheroidal coordinates are built from exactly two foci; a third nucleus has nowhere to sit. Gaussians escape this because two Gaussians on different centers multiply into a single Gaussian on a third point [31]—Slater orbitals have no product theorem—and this, not any Geometric-Vacuum-specific limitation, is the seventy-year-old reason Gaussian bases displaced Slater orbitals for polyatomic quantum chemistry. The modern state of the art for three-center Slater repulsion integrals in linear conformations (Ref. [7]) reduces them to a single infinite expansion or a one-dimensional numerical quadrature; the one celebrated closed form in correlated three-electron theory, Fromm and Hill's analytic evaluation [9], is a *dilogarithm*—a genus-zero object—and belongs to a different integral topology (the one-center, explicitly correlated master integral), which lacks the structure we are about to exhibit.

This paper asks what the three-center integral *is*, transcendentally, and answers for its radial core: an elliptic object. The two-center weight-one closure is a genus-zero fact and it stops at two centers. The third center raises the radial core's transcendence one genus, to the elliptic (genus-one) world of Bessel moments and two-mass sunrise Feynman integrals. We establish this constructively, validate every step numerically, place the object in the literature, and mark the finite closed form as open. As a function of the bond angle, the integrated value also leaves the two-center class (Sec. VIII).

Lineage and scope. The momentum-space route is not new machinery. Molecular integrals over exponential-type orbitals have long been evaluated by Fourier transformation, molecular orbitals in momentum space going back to Shibuya and Wulfman [6]: the transform of a two-center orbital product is classical [33], its reduction to four-dimensional (Fock) hyperspherical harmonics goes back to Niukkanen [34], and the general one- and two-electron multicenter integrals close, via the Fourier-transform method together with Feynman's identity, to low-dimensional numerical quadrature [35]. Recently the four-center two-electron integral over Slater orbitals has been reduced, by cached Fourier–Bessel coefficients, to one radial quadrature, at about 4×10^5 integrals per second on one GPU at the contraction stage, with the cache already built, and about 4×10^4 per second end to end, in a preliminary benchmark [10]; that is the production baseline for any cost claim below. Formulas in elementary functions for three- and four-center Slater integrals have also been given [11], but they drop the cross term in the gradients of two orbitals, on the ground that the cosine between them averages to zero, so they are estimates, not exact closed forms. Momentum-space Coulomb–Sturmians are hyperspherical harmonics under Fock's conformal projection [2], the framework the Averys built into the generalized-Sturmian method [3, 4, 43] and applied to Coulomb–Sturmian repulsion integrals [5]. The closest prior art to the present calculation is Avery's momentum-space route [42]—the same Fock-projection method on the same objects, and the one place the elliptic structure was nearest to visible; a targeted search finds it, like all prior work, carrying the evaluation to expansion and numerics without drawing the genus reading, though this is a search-negative inferred from the published abstracts, not a verified absence. What is new here is the reading: that the residual radial object is a specific elliptic Bessel moment, and hence that a molecular integral is an instance of the elliptic-Feynman-integral class. We work throughout with $1s$ orbitals at a common exponent; the generalization to higher (n, ℓ, m) multiplies the integrand by solid harmonics and shifts exponents but does not change the genus (Paper 63 [1] proves this for every exponential-type orbital), and we return to this in Sec. VIII.

II. THE MOMENTUM-SPACE REDUCTION

Write the chemist integral for two densities $\rho_1 = \chi_X\chi_Y$ and $\rho_2 = \chi_X\chi_Z$, two two-center $1s$ densities that share the center X on different axes. With the Fourier representation of the kernel, $1/|\boldsymbol{r}_1 - \boldsymbol{r}_2| = (2\pi^2)^{-1}\int d^3k\, e^{i\boldsymbol{k}\cdot(\boldsymbol{r}_1-\boldsymbol{r}_2)}/k^2$, the integral becomes

$$(XY|XZ) = \frac{1}{2\pi^2}\int \frac{d^3k}{k^2}\,\tilde\rho_1(\boldsymbol{k})\,\overline{\tilde\rho_2(\boldsymbol{k})}, \qquad \tilde\rho(\boldsymbol{k}) = \int \rho(\boldsymbol{r})\, e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\, d^3r. \tag{1}$$

Each center now appears only through a phase $e^{i\boldsymbol{k}\cdot\boldsymbol{R}}$: three centers are three phases, not three foci, and there is nothing to diagonalize. Equation (1) is exact; the only content is the density Fourier transform $\tilde\rho$ and the $\boldsymbol{k}$ integral.

We verify Eq. (1) two independent ways on the reference geometry $X = (0,0,0)$, $Y = (0,0,2)$, $Z = (1.5,0,0.5)$ (bohr), $\zeta = 1$, for which a McMurchie–Davidson evaluation over 8-Gaussian $1s$ fits gives the reference value $(XY|XZ) = 0.204941722$. That value carries the fit error: as the number of Gaussians grows (6 to 16) the evaluation converges to the Slater value 0.2049416525, 7×10^{-8} lower, which an exact expansion in angular channels about the shared center reproduces to 3×10^{-10} (`tests/test_pairwise_three_center.py`). **[MEASURED]** Using the closed-form Fourier transform of the Gaussian-fitted density (a product of Gaussians is a Gaussian) in Eq. (1) reproduces the McMurchie–Davidson value on the *same* fits to 1.9×10^{-14}: the momentum formula, the Fourier and Coulomb conventions, and the $\boldsymbol{k}$-quadrature are correct. Independently, computing the *true* Slater density transform (no Gaussian fit) by a one-parameter reduction of the

two-center Yukawa product reproduces that reference to 4.3×10^{-7}. Both are pinned in the regression suite [54].

The angular integral closes to a spherical Bessel kernel. Writing each Slater factor through $e^{-\zeta r} = -\partial_\zeta(e^{-\zeta r}/r)$ and the Feynman parametrization of the resulting Yukawa product, the three phases combine into a *single* phase $e^{i\boldsymbol{k}\cdot\boldsymbol{W}}$ with $\boldsymbol{W}(s,t) = (t-s)\boldsymbol{X} + s\boldsymbol{Y} - t\boldsymbol{Z}$ (s, t the two Feynman parameters), so the angular integral is elementary,

$$\int d\Omega_{\boldsymbol{k}}\, e^{i\boldsymbol{k}\cdot\boldsymbol{W}} = 4\pi\, j_0(k|\boldsymbol{W}|) = 4\pi\, \frac{\sin(k|\boldsymbol{W}|)}{k|\boldsymbol{W}|}. \tag{2}$$

The whole integral reduces to a two-dimensional Feynman integral over a one-dimensional radial (k) integral,

$$(XY|XZ) = \frac{8}{\pi}\, \partial_{\zeta_a}\partial_{\zeta_b}\partial_{\zeta_c}\partial_{\zeta_d} \int_0^1 ds \int_0^1 dt \int_0^\infty dk\; j_0(k|\boldsymbol{W}|) \times \frac{e^{-D_1\Delta_1}}{\Delta_1}\, \frac{e^{-D_2\Delta_2}}{\Delta_2}\, \Bigg|_{\zeta=1}, \tag{3}$$

with $\Delta_1 = \sqrt{c_1 k^2 + 1}$, $\Delta_2 = \sqrt{c_2 k^2 + 1}$, $c_1 = s(1-s)$, $c_2 = t(1-t)$, and $D_1 = |X - Y|$, $D_2 = |X - Z|$. The ∂_ζ's (a bookkeeping device restoring the Slater from the Yukawa) are elementary operations; the transcendental content enters through the radial factor, and the (s, t) integrations can lower its class (Sec. VII). Equation (3) reproduces the 8-Gaussian reference to 1.8×10^{-6} **[MEASURED]**, the residual being the finite-order angular and Feynman quadrature.

III. THE TRANSCENDENCE JUMP: GENUS ZERO TO GENUS ONE

The radial factor of Eq. (3) carries two dispersion factors Δ_1, Δ_2 with *different* scales $c_1 \neq c_2$. A single such factor is benign. Under Fock's substitution $k\sqrt{c} = m \sinh\theta$ it linearizes, and **[MEASURED,** 6×10^{-18}**]**

$$\int_0^\infty \cos(kb)\, \frac{e^{-D\sqrt{c\,k^2+m^2}}}{\sqrt{c\,k^2+m^2}}\, dk = \frac{1}{\sqrt{c}}\, K_0\Big(\tfrac{m}{\sqrt{c}}\sqrt{c\,D^2+b^2}\Big), \tag{4}$$

a Bessel K_0: the momentum-space Coulomb–Sturmian object, living on a rational (genus-zero) curve. This is why the two-center engine is weight one.

Two different scales cannot be linearized together. Their product is governed by the algebraic curve

$$y^2 = (c_1 k^2 + 1)(c_2 k^2 + 1), \tag{5}$$

a quartic with four distinct branch points $k = \pm i/\sqrt{c_1}, \pm i/\sqrt{c_2}$ whenever $c_1 \neq c_2$: an *elliptic curve*, genus one. When $c_1 = c_2$ the branch points collide and the quartic becomes a perfect square—rational, genus zero. The cleanest witness is the period. At $D_1 = D_2 = 0$ the radial integral is exactly a complete elliptic integral **[MEASURED, 31 digits; classical]**,

$$\int_0^\infty \frac{dk}{\sqrt{(c_1k^2+1)(c_2k^2+1)}} = \frac{1}{\sqrt{c_{\max}}}\, K(m), \tag{6}$$

with modulus $m = 1 - c_{\min}/c_{\max}$ (nondegenerate, $0 < m < 1$; prefactor verified for both orderings of c_1, c_2 in `test_paper59_euclidean_dictionary.py`; in Bessel form it is Eq. (19) of Ref. [14]), while on the diagonal $c_1 = c_2 = c$ it collapses to the elementary $\pi/(2\sqrt{c})$. Since $c_1 = s(1-s)$ and $c_2 = t(1-t)$ range independently over $[0, \frac{1}{4}]$, the full integral (3) is an integral over a *family* of elliptic curves, of modulus $m(s,t) = 1 - c_{\min}/c_{\max}$, degenerate only on the measure-zero locus $s = t$ or $s = 1-t$. Every term generated by the ∂_ζ's is a meromorphic function on the same curve (5), so the whole integral is an integral of periods over this elliptic family.

The structural statement is that the embedding tier of the transcendence taxonomy (Papers 18, 34 [50, 51]) is *genus-graded* at the level of the radial core: genus zero ($\{E_1, \ln, \gamma\}$) at two centers, genus one (elliptic) at three. The two two-center negatives—no π, no dilogarithm—are genus-zero properties; at the third center the radial core does not climb to a higher rung of the polylogarithm weight tower but leaves that tower for the elliptic one. This is the transcendence-level form of "no coordinate system for three foci." It does not by itself fix the class of the integrated value: the two-center exchange class builds the same fibre family and still closes at genus zero (Sec. VII). The integrated three-center value is tested separately, as a function of the bond angle, and it leaves the two-center class (Sec. VIII).

IV. THE REDUCTION: AN ELLIPTIC BESSEL MOMENT

The radial factor can be named precisely. Applying Eq. (4) to each factor and Parseval's theorem, the radial core (at fixed Feynman parameters, before the elementary j_0 and ∂_ζ wrapping) is a two-mass Bessel moment **[MEASURED, 30 digits]**,

$$M = \frac{2}{\pi\sqrt{c_1 c_2}} \int_0^\infty db\; K_0\big(a_1\sqrt{p_1^2+b^2}\big)\, K_0\big(a_2\sqrt{p_2^2+b^2}\big), \tag{7}$$

with $a_i = 1/\sqrt{c_i}$ and $p_i = D_i\sqrt{c_i}$. This is a shifted-argument, two-scale member of the Bessel-moment family; it is not one of the moments catalogued in closed form by Bailey, Borwein, Broadhurst and Glasser [14], whose two-scale entries (their Eqs. 19 and 30) are the unshifted case $p_1 = p_2 = 0$.

A change of variable $x = \sqrt{c_1 k^2 + 1}$ on the one-mass slice $N(D) \equiv M(D, 0)$ puts it in the cleanest canonical form: the Laplace transform of the holomorphic elliptic differential,

$$N(D) = \frac{1}{\sqrt{c_1}} \int_1^\infty \frac{e^{-Dx}\, dx}{\sqrt{(x^2-1)(\rho\, x^2 + 1 - \rho)}}, \qquad \rho = \frac{c_2}{c_1}. \tag{8}$$

This is a single explicit one-dimensional integral of an elementary integrand. A low-dimensional reduction is the expected output of the Fourier-transform method—the same machinery reduces multicenter Slater integrals to two- or three-dimensional quadrature [35] and three-center integrals to a semi-infinite oscillatory Bessel integral [36]—so what is new is not the reduction itself but the *elliptic reading* of the integrand it produces, which makes the genus-one structure manifest. It is an *elliptic deformation of the Bessel* K_0: as $\rho \to 0$ (the one-mass limit, one scale decoupling) $N \to K_0(D)$, and at $D = 0$ it is the complete elliptic integral K of Eq. (6).

Because $1/\sqrt{Q}$, $Q(x) = (x^2-1)(\rho x^2 + 1 - \rho)$, satisfies the first-order relation $Qf' + \frac{1}{2}Q'f = 0$, and the boundary at the branch point $x = 1$ vanishes ($Q(1)^{1/2} = 0$), integration by parts against e^{-Dx} (using $\int_1^\infty x^n e^{-Dx} f\, dx = (-1)^n L^{(n)}(D)$, $L \equiv \sqrt{c_1}\, N$) yields an exact fourth-order Picard–Fuchs equation with polynomial coefficients **[SYMBOLIC PROOF]**,

$$D\rho\, L'''' + 2\rho\, L''' + D(1-2\rho)L'' + (1-2\rho)L' - D(1-\rho)L = 0. \tag{9}$$

Its characteristic polynomial factors as $\rho(s^2-1)\big(s^2 + (1-\rho)/\rho\big)$, whose roots $s = \pm 1$ and $s = \pm i\sqrt{(1-\rho)/\rho}$ are exactly the four branch points of the curve (5): the K_0/I_0 sector from the (x^2-1) pair, the oscillatory J_0/Y_0 sector from the other. (A differential-equation method for molecular two-electron integrals is itself established: Pachucki [41] evaluates the *two*-center integral from a second-order equation in the internuclear distance, closing it in elementary functions; the genus jump appears here as the step from that second-order, genus-zero equation to the fourth-order Picard–Fuchs equation of an elliptic curve.) A subordination identity casts the same object as a two-scale sunrise integral, $\propto \iint ds_1 ds_2\, (s_1 s_2)^{-1/2} e^{-D_1^2/4s_1 - s_1 - D_2^2/4s_2 - s_2}/\sqrt{s_1 c_1 + s_2 c_2}$, whose second-Symanzik-type $\sqrt{\text{linear form}}$ is the genus-one signature and separates only when $c_1 = c_2$.

One numerical fact pins the object's class beyond the period. $N(D)$ is *non-analytic in D at $D = 0$*: it carries a $D \ln D$ term **[MEASURED]**, so it is not a pure elliptic integral (K/E are analytic in their parameters). The elliptic $D = 0$ period together with the $D \ln D$ non-analyticity are the two fingerprints of an *elliptic ⊕ logarithmic* object—the inhomogeneous ("regulator") term of the Picard–Fuchs solution associated with the elliptic dilogarithm [20, 21, 44]—a correspondence whose reach we test, and bound, in Sec. V.

[SYMBOLIC + MEASURED] *The kinematics.* Equation (4) is the frequency representation of the two-dimensional Euclidean propagator: with $G_a(\tau, x) = (2\pi)^{-1} K_0(a\sqrt{\tau^2 + x^2})$ the Green's function of $-\Delta + a^2$ on $\mathbb{R}^2$, its left-hand side is exactly $(2\pi/\sqrt{c})\, G_a(p, b)$ with mass $a = \zeta/\sqrt{c}$, Euclidean time $p = D\sqrt{c}$ and space b, and the fibre (7) is exactly the two-propagator correlator $M = 4\pi a_1 a_2 \int_{\mathbb{R}} db\, G_{a_1}(p_1, b)\, G_{a_2}(p_2, b)$: two Euclidean packets of unequal mass, evolved from a point source for Euclidean times p_1, p_2 and overlapped on one spatial slice **[MEASURED, 10^{-30} and 10^{-31} at five parameter points each; regression-tested at one point each, to 10^{-18} and 10^{-14}, in `test_paper59_euclidean_dictionary.py`]**. Three things become visible. First, the Feynman parameters are Källén–Lehmann spectral variables: $c = s(1-s)$ forces $a = \zeta/\sqrt{c} \geq 2\zeta$, so each two-centre density is dispersed over its own two-particle cut and the elliptic modulus $\rho = c_2/c_1 = (a_1/a_2)^2$ is the squared ratio of the two spectral masses—the genus jump is a mass-splitting phenomenon, and the three cusps are equal masses ($\rho = 1$) and one mass at the heavy end of its cut ($\rho \to 0, \infty$). Second, the exponential twist is $a_i p_i = \zeta D_i$, *independent of the Feynman parameter*: the family is swept at fixed mass×time, and the physical point is one Compton wavelength of Euclidean-time displacement, the $u = D$ endpoint of the source-time superposition $e^{-D\Delta}\Delta^{-n} = \frac{1}{(n-1)!}\int_D^\infty (u-D)^{n-1} e^{-u\Delta}\, du$ that converts the Yukawa into the Slater. Third, $j_0(k|\boldsymbol{W}|)$ is a uniform smearing of the shared spatial coordinate over the composite separation, so the third centre enters *only* through space while the bond lengths set the two Euclidean-time offsets **[MEASURED, 1.8×10^{-10} for the full position-space fibre; not regression-tested]**.

V. THE ELLIPTIC-DILOGARITHM ROUTE IS OBSTRUCTED

The unequal-mass sunrise is closed by variation of parameters: a second-order Picard–Fuchs operator in the external variable annihilates the two periods, its inhomogeneity is an elementary subtopology (a tadpole), and the particular solution—an iterated integral of the periods against that elementary source—is the elliptic dilogarithm [20, 21]. (The full unequal-mass sunrise has since been closed to all orders in ε as iterated integrals on $\overline{\mathcal{M}}_{1,3}$ by Bogner, Müller-Stach and Weinzierl [22]—a *Fuchsian* object; the point below is that $N(D)$ is its irregular Laplace dual and so lies outside that technology.) Running the same machinery on $N(D)$ is the natural attempt. It does not close, and the reason is structural.

The curve $y^2 = Q(x, \rho)$ of Eq. (5) moves with the modulus ρ, not with the Laplace variable D. Accordingly the two periods $K(\rho), K(1-\rho)$ are the homogeneous solutions of the second-order modulus operator **[MEASURED: both annihilated to 10^{-25}; $L(0, \rho) = K(1-\rho)$ to 19 digits]**

$$\mathcal{M}_\rho = \rho(1-\rho)\,\partial_\rho^2 + (1-2\rho)\,\partial_\rho - \tfrac{1}{4}, \tag{10}$$

and they are *not* solutions of the fourth-order operator in D: a constant gives $\mathcal{O}[1] = -D(1-\rho) \neq 0$, and the single Bessel functions of both sectors, $K_0(D), I_0(D)$ and $J_0(\omega D), Y_0(\omega D)$ with $\omega = \sqrt{(1-\rho)/\rho}$, are likewise not D-solutions **[MEASURED]** (they are coupled, which is why Eq. (9) does not factor into elementary Bessel operators). The elliptic-dilogarithm construction lives on the periods, so it must be run in the modulus ρ.

There it meets the obstruction. Acting with $\mathcal{M}_\rho$ under the integral sign is an exact total derivative **[SYMBOLIC PROOF; exact residual** 0**]**,

$$\mathcal{M}_\rho\big[Q^{-1/2}\big] = \frac{d}{dx}\Big[\, g(x)\, Q^{-1/2}\Big], \quad g(x) = \frac{-x(x^2-1)}{4(\rho x^2+1-\rho)}, \tag{11}$$

so for $D \neq 0$ the modulus source is $S(D,\rho) \equiv \mathcal{M}_\rho[L] = D\int_1^\infty e^{-Dx} g\, Q^{-1/2} dx$. Because g *vanishes at the branch point* $x = 1$, differentiating in the modulus produces no branch-point boundary term: the analogue of the sunrise's pinched propagator—the elementary tadpole that would be the inhomogeneity—is absent, and the source stays on the elliptic curve. Quantitatively **[MEASURED]**, S closes exactly and finitely only inside the integral's own Gauss–Manin D-module,

$$S(D,\rho) = \sum_{k=0}^{3} P_k(D,\rho)\, \partial_D^k L, \qquad \deg_D P_k = 2, \tag{12}$$

with held-out residual saturating flat under increase of the fit degree (below 10^{-25} in the regression witness, 5×10^{-44} at driver precision; numerically an exact finite identity); whereas the genuine variation-of-parameters shape—period level $\{L, \partial_D L\}$ plus an elementary Bessel subtopology $\{K_0, K_1\}(D)$, $\{J_0, Y_0\}(\omega D)$—only *approximates* it (residual 2×10^{-14} falling geometrically to 3×10^{-30} across fit degrees, never saturating). The inhomogeneity is in-module, not a simpler subtopology, and variation of parameters against the periods is therefore circular.

[OBSERVATION] $N(D)$ is thus one transcendence level above the sunrise—above the *fully solved* sunrise, not merely its leading elliptic dilogarithm: the all-orders unequal-mass sunrise is now closed as iterated integrals on $\overline{\mathcal{M}}_{1,3}$ [22], a period of a *Fuchsian* connection, whereas $N(D)$ is not. It is not an elliptic dilogarithm of elementary or period-level arguments; it is a period of the rank-four irregular connection defined by Eq. (9), for which the sunrise mechanism supplies no reduction to the elliptic-polylogarithm tower [47]. **[SYMBOLIC]** The irregularity is explicit at the operator level: at $D = \infty$ the operator of Eq. (9) admits four formal exponential solutions $e^{\lambda D}$ with $\lambda \in \{\pm1, \pm i\sqrt{(1-\rho)/\rho}\}$—its leading Newton edge is a full-width slope-one segment, so it is irregular of Poincaré rank one, the four λ being exactly the branch points (the K_0/I_0 and J_0/Y_0 Bessel sectors), whereas the sunrise/Legendre period operator is regular singular at infinity ($\lambda = 0$ only). The two operators therefore sit on opposite sides of the Laplace transform, and the Fuchsian elliptic-polylogarithm machinery of [22] cannot reach $N(D)$. Whether $N(D)$ admits a closed form of any kind is a differential-Galois question—whether the fourth-order operator factors over the differential field generated by the periods $K(\rho), K(1-\rho)$—settled below, negatively, at the reducibility level (the operator does not factor); what remains open is only whether a *transcendental* closed form exists at the irreducible level.

[MEASURED] The same irregularity is visible in $N(D)$ itself, as a *resurgent* structure. Expanded about the Bessel point $\rho = 0$ ($N \to K_0(D)$), the one-mass slice has the explicit elementary term series

$$g(1) = 0, \qquad N(D) = \frac{1}{\sqrt{c_1}} \sum_{n\geq0} (-1)^n \frac{\big((2n-1)!!\big)^2}{2^n\, n!}\, \rho^n\, \frac{K_n(D)}{D^n}, \tag{13}$$

each coefficient a rational multiple of an *integer-order* Bessel function—the transcendence does not escalate—but the series is *asymptotic*, diverging factorially (optimal truncation at $n \sim 1$–3, a few digits, then unbounded). Equivalently the $D \to \infty$ expansion $N(D) \sim e^{-D}\sum_k b_k D^{-(k+1/2)}$ is Gevrey-one, and its Borel transform is the algebraic function $\psi(\zeta) = [(\zeta+2)(\rho(1+\zeta)^2 + 1-\rho)]^{-1/2}$ on the very curve (5), singular exactly at the three non-dominant branch points $\zeta \in \{-2, -1 \pm i\sqrt{(1-\rho)/\rho}\}$—the Stokes actions linking the e^{-D} sector to the e^{+D} and $e^{\pm i\sqrt{(1-\rho)/\rho}\, D}$ ones. A Borel–Padé resummation confirms these locations (at $\rho = \frac{1}{5}$: $\zeta = -2$ and $-1 \pm 2i$, recovered to $\sim 1\%$). So the residual transcendence of $N(D)$ is not exotic: it is the resurgence of an elementary Bessel series, its Stokes data pinned to the branch points of the elliptic curve.

[MEASURED] The resurgence, moreover, *closes with algebraic Stokes data.* The Borel transform $\psi(\zeta)$ carries only square-root singularities, so its amplitude at each is algebraic: $a_* = 1$ at $\zeta = -2$, and $a_*^2 = -\frac{1}{2} \mp \frac{i}{2}\sqrt{\rho/(1-\rho)}$ at $\zeta = -1 \pm i\sqrt{(1-\rho)/\rho}$ (matched to the closed-form residue to 10^{-37}, and confirmed to be the large-order growth constants by singularity analysis; the dominant Stokes ray switches from the real to the complex pair at $\rho = \frac{1}{4}$). These amplitudes *are* the Stokes constants connecting the four exponential sectors, so the entire trans-series of $N(D)$ over $\{e^{\pm D}, e^{\pm i\sqrt{(1-\rho)/\rho}\, D}\}$ has *algebraic* connection data. The one genuinely elliptic ingredient enters only at the boundary: at $D = 0$ the open Laplace thimbles close into the cycles of the curve and $\sqrt{c_1}\, N(0) = K(1-\rho)$ exactly (14 digits). So the representation at the irregular level is explicit: $N(D)$ is the Borel–Laplace transform of the algebraic differential ψ, with algebraic Stokes data and an elliptic-period $K(1-\rho)$ normalization—as far as a genuinely irregular, genus-one object reduces. What remains transcendental is the elliptic period at the boundary and the standard Borel–Laplace resummation, not any new constant.

[OBSERVATION] Structurally this is the genus-one analogue of the classical resurgence of the modified Bessel function K_0—the archetypal rank-one irregular equation, whose large-argument series has an algebraic Borel transform and elementary Stokes data [23, 24]. The natural home for the algebraic-Stokes-versus-transcendental-period dichotomy seen here is Sabbah's theory of Stokes structures for Fourier–Laplace transforms of algebraic connections [25], in which the Stokes data is local at the irregular point (hence algebraic) while transcendental periods enter through the regular points (hence $K(1-\rho)$

at $D = 0$). The closest worked instance is the rank-two, genus-*zero* Bessel/Kloosterman connection treated via irregular Hodge theory [29]; and the only resurgence analysis of the sunrise/banana family we found operates in the modular parameter $q = e^{2\pi i\tau}$ (the $|q| \to 1$ limit), not the Laplace-dual D [26]. The present genus-one, Laplace-dual reading appears unmade; an absence from a targeted search, not a proof.

A concrete route from the solved dual. Because $N(D)$ is the Laplace transform of the sunrise elliptic differential and the unequal-mass sunrise is now known in full closed form [22], one concrete—if not obviously tractable—route to $N(D)$ is to Laplace-transform that solution directly, an operation that necessarily leaves the elliptic-polylogarithm class for the Bessel-moment/resurgence one, consistent with the irregular character just established. We have not carried this out.

The rank-four label is not vacuous: a base-field reducibility scan. That $N(D)$ is a period of a genuinely rank-four object—rather than one secretly carrying lower-rank content—is itself a reducibility statement, and it can be tested directly. Two structural facts constrain Eq. (9). First, the operator is *formally self-adjoint*, $L^* = L$ **[SYMBOLIC PROOF]**, so its differential Galois group lies in the symplectic group $\mathrm{Sp}_4(\mathbb{C})$—the signature of a polarized period. Second, its indicial exponents at the regular singular point $D = 0$ are $\{0, 1, 1, 2\}$ **[SYMBOLIC PROOF]**; the *double* exponent at 1 is the local-monodromy source of the $D \ln D$ nonanalyticity noted above. A search over the base field $\mathbb{Q}(\rho)(D)$ then returns *no* factor of order one or two—hence, by self-adjointness, none of order three—with polar locus at the true singularities $\{0, \infty\}$ **[SYMBOLIC, positive-control validated]**: there is no hyperexponential solution $e^{\lambda D}Q(D)$ for any $\lambda \in \{\pm 1, \pm i\omega, 0\}$, and the only two order-two factorizations the symbol $(s^2 - 1)(s^2 + \omega^2)$ admits over $\mathbb{Q}(\rho)$—the two Bessel sectors—both fail. The two natural elementary structures are excluded by the exponents alone: L is neither the symmetric product of its Bessel sectors (which would carry exponents $\{0, 0, 0, 0\}$) nor a symmetric cube. This is strong evidence that Eq. (9) is irreducible over $\mathbb{Q}(\rho)(D)$, and hence that the rank-four label is earned. The eigenring sharpens it to a single residual target. The endomorphism algebra of the solution space is trivial, $E(L) = \mathbb{C}$ **[computed at $\rho = 1/5$; robust across pole windows]**, so L is indecomposable; and the symplectic form pairs the exponential solution y_λ at $D = \infty$ with $y_{-\lambda}$. A non-degenerate invariant subspace would split $V = W \oplus W^\perp$ and enlarge the eigenring, so every invariant subspace is *isotropic*; the exponential torus $T = (\mathbb{C}^*)^2$ confines invariant subspaces to spans of the four lines $\{y_{\pm 1}, y_{\pm i\omega}\}$, whose two symplectic pairs (the Bessel sectors) are non-degenerate. A one-dimensional isotropic invariant subspace is a hyperexponential solution, already excluded. Hence L is reducible over $\mathbb{C}(D)$ *iff* one of the four "mixed" Lagrangian planes $\mathrm{span}\{y_{\pm 1}, y_{\pm i\omega}\}$ is invariant—equivalently, iff the Stokes matrices at $D = \infty$ [25] fail to couple the real exponential pair $\{\pm 1\}$ to the imaginary pair $\{\pm i\omega\}$. This is decidable, and it is settled twice over—once in principle, once explicitly.

In principle. The reduction is an instance of the Fourier–Laplace transform of D-modules, where irreducibility is automatic. Let $\mathcal{L}$ be the rank-one meromorphic connection on $\mathbb{P}^1$ whose flat sections are $Q^{-1/2}$: it is regular singular, with local monodromy -1 at each of the four branch points $x \in \{\pm 1, \pm i\omega\}$ and trivial monodromy at ∞ (there $Q^{-1/2} \sim x^{-2}$). Having rank one, $\mathcal{L}$ is irreducible, so its minimal (intermediate) extension $j_{!*}\mathcal{L}$ to $\mathbb{A}^1$ is a *simple* holonomic D-module. (Because every finite local monodromy is $-1 \neq 1$ and hence has no invariant, $j_!\mathcal{L} = j_{!*}\mathcal{L} = j_*\mathcal{L}$, so "the D-module generated by $Q^{-1/2}$" is unambiguous.) The Laplace kernel e^{-Dx} makes the radial integral the Fourier–Laplace transform $\mathrm{FL}(j_{!*}\mathcal{L})$, and FL is an exact auto-equivalence of the category of holonomic D-modules on the affine line (with its dual coordinate) [30]; an equivalence sends simple objects to simple objects, so $\mathrm{FL}(j_{!*}\mathcal{L})$ is simple. Its generic rank is the total drop of $\mathcal{L}$—one unit at each branch point, where the monodromy -1 leaves no invariant, and none at the regular point ∞, hence *four*, matching the order of Eq. (9), of which $N(D)$ is the defining (cyclic) solution. A simple holonomic D-module of generic rank four restricts to an irreducible connection over the function field, so **the operator of Eq. (9) is irreducible** over $\mathbb{C}(D)$, hence over $\mathbb{Q}(\rho)(D)$. Simplicity is a property of the connection at the generic point, so it forbids a factor of *any* order—apparent singularities and all; the monodromy computation next verifies it by hand.

Explicitly. The same conclusion is exhibited, and the mechanism made visible, by the monodromy. The four exponentials are the branch points $x \in \{\pm 1, \pm i\omega\}$ of $y^2 = Q$ in the Borel (Laplace-dual) plane, and the monodromy of L around $D = 0$, computed in the exponential-line basis of Lefschetz-thimble solutions $\int e^{-Dx}\, dx/\sqrt{Q}$, is **[MEASURED; exact-integer certificate]** the integer unipotent matrix

$$M_0 = \begin{pmatrix} -1 & 2 & 2 & 2 \\ -2 & 3 & 2 & 2 \\ -2 & 2 & 3 & 2 \\ 2 & -2 & -2 & -1 \end{pmatrix},$$

the *same* for $\rho = \frac{1}{5}$ and $\rho = \frac{1}{3}$ (a topological invariant of the branch-point configuration), with a single 2×2 Jordan block—the lone $D \ln D$ log—and *no* invariant coordinate subspace. The Stokes matrices therefore *do* couple the real and imaginary exponential pairs, no mixed plane survives, and the reducibility criterion above returns *irreducible*—confirming the structural argument concretely. **[SYMBOLIC + MEASURED]** Consequently $N(D)$ is a period of an *irreducible* rank-four connection: it admits no closed form through any factorization of L, and the sole remaining route to a closed form is a genuinely transcendental one at the full rank-four level—an Eisenstein / complex-multiplication Bessel-moment period of the Legendre ($\Gamma(2)$) family, of which a $\Gamma(2)$ multiple

modular value is only the regular $D \to 0$ shadow (Paper 61 [53]). The differential-Galois *reducibility* question is thereby settled (negatively: L is irreducible); the transcendental closed form is the one problem that remains open [54].

VI. PLACEMENT IN THE LITERATURE

The object of Eqs. (7)–(9) sits inside a fully developed body of mathematics, applied so far only to problems far from chemistry. Two-mass elliptic Feynman integrals—the sunrise/banana family—are evaluated in terms of elliptic dilogarithms as inhomogeneous solutions of exactly this kind of Picard–Fuchs equation [20, 21, 27]; their periods are the complete elliptic integrals of Eq. (6), and at special (complex-multiplication) scale ratios their values reach critical L-values of modular forms [28, 29, 48, 49]. The closest published analog to Eq. (7) is a doubled-argument Bessel moment that Broadhurst evaluated as a product of complete elliptic integrals via a lattice-Green-function contour [15]: the same "two-scale moment $\to$ product of K's" mechanism we expect at rational ρ.

On the chemistry side the treatment is uniformly numerical: three-center Slater repulsion integrals are reduced to infinite series or one-dimensional quadrature [7, 8], and the Averys' momentum-space Coulomb–Sturmian route [5, 42], closest in spirit to ours, likewise carries the evaluation to expansion and numerics with no genus reading our search could find. We could find no paper drawing the bridge in either direction: no chemistry reference names an electron-repulsion integral as a Bessel moment or an elliptic Feynman integral, and no amplitudes reference applies the elliptic machinery to a molecular integral. **[OBSERVATION]** The identification—that the three-center Slater integral, via the Fock projection, is a two-scale Bessel moment on the sunrise elliptic curve—appears to be unmade. This is an absence from a targeted search, not a proof of absence; the residual risk is an obscure single source. **[OBSERVATION]** A second placement datum is arithmetic: the $\mathbb{Z}[i]$ content this family carries (the theta series θ_3^2 of the companion (Paper 61); $G = \beta(2) = L(2, \chi_{-4})$) does not appear in the established Bessel-moment corpus, whose forms sit at levels $3, 6, 8, 15$ [28, 29, 48] and, in the twisted moments of Lim, Tu and Yu [46], which are the moments $c_{n,0}$ of the tree ladder (Paper 63 [1]), 24 and 48: a targeted search finds no Catalan-constant appearance in any of these, and χ_{-4} enters $c_{3,0}$ as a twist: the form attached to it is the newform of level 12 twisted by χ_{-4}, whose complex multiplication stays by $\mathbb{Q}(\sqrt{-3})$, not $\mathbb{Q}(i)$ [45]; the nearest other χ_{-4} appearance is the topological-string sector of the Broadhurst–Dorigoni programme [26]. The family thus occupies an apparently unclaimed arithmetic slot ($\mathbb{Z}[i]$, complex multiplication by $\mathbb{Q}(i)$) in that literature, of a piece with the unmade bridge above. (Reconciliation with the value-level searches of the companion (Paper 61): the conductor-4 arithmetic is a property of the *family* and of the $D \to 0$ shadow; the physical integrated value itself shows no measured conductor-4 content at reachable heights.)

The modular and complex-multiplication structure of this elliptic family, its cosmic-Galois periods, and the Bessel-moment period algebra of the object—the master family, the Wronskian $W_0 D^{-2}$, and the quadratic period relations $B = \pi\,\Omega$ with Ω the canonical integer symplectic intersection form—are developed in the companion Paper 61 [53], which takes the present reduction as its starting point.

VII. EXPLICITLY-CORRELATED INTEGRALS ARE NATIVE TO THIS REPRESENTATION

The momentum reduction of Sec. II is *kernel-agnostic*. A two-electron matrix element of *any* operator $f(r_{12})$ between orbital products is, by the convolution theorem,

$$\langle \phi_a \phi_b \,|\, f(r_{12}) \,|\, \phi_c \phi_d \rangle = \frac{1}{(2\pi)^3} \int d^3k\; \widetilde{\rho}_{ac}(\boldsymbol{k})^*\, \widetilde{f}(\boldsymbol{k})\, \widetilde{\rho}_{bd}(\boldsymbol{k}), \qquad \rho_{ac} = \phi_a \phi_c \tag{14}$$

the identical object as the electron-repulsion integral with the Coulomb kernel $\widetilde{(1/r_{12})} = 4\pi/k^2$ replaced by $\widetilde{f}$, and the angular closure to the spherical Bessel kernel of Eq. (2) applies unchanged. Explicitly-correlated (F12) methods take f to be a correlation factor; the standard choice is the Slater geminal $f = e^{-\gamma r_{12}}$, whose transform $\widetilde{f}(k) = 8\pi\gamma/(k^2+\gamma^2)^2$ is exact, elementary, and—unlike the Coulomb kernel—*regular* at $k = 0$. In this representation the geminal is therefore a native object: no expansion of $e^{-\gamma r_{12}}$ in Gaussians (the first workaround of Gaussian-basis F12) is required. **[MEASURED]** The momentum form (14) reproduces a direct position-space quadrature to 10^{-14}, and the Coulomb self-repulsion of a $1s$ density to its exact value $5a/8$ (`test_f12_kernel_swap_and_native_geminal`).

At the level of the curve, whether the integral closes in elementary functions is governed *not* by the operator but by the *density topology*; the class of the integrated object can still move with the kernel's mass and shape (the two-center master integral below, and the correlated triangle of Paper 63 [1]). **[SYMBOLIC]** When both transition densities are single-center the radial integrand is rational and the pair integral is genus zero: e.g. $\int_0^\infty [k^2/(k^2+\gamma^2)^2]\, j_0(kR)\, dk = (\pi/4\gamma)\, e^{-\gamma R}$ (`test_f12_geminal_pair_is_elementary`). A *two*-center transition density, however, contributes a Fock dispersion factor $\sqrt{c\,k^2+1}$ (Sec. IV), and two of them build the very curve of Eq. (5), $y^2 = (c_1 k^2+1)(c_2 k^2+1)$, whose period is a complete elliptic integral (`test_f12_multicenter_shares_eri_elliptic_genus`). A rational kernel—Coulomb $4\pi/k^2$ or geminal $8\pi\gamma/(k^2+\gamma^2)^2$ alike—is a rational function on this curve, so $\widetilde{f}\,dk/y$ is again an elliptic integral: the genus is *unchanged*, the geminal merely trading an elliptic integral of the first kind (the Coulomb period) for one

of the third kind (its poles at $k^2 = -\gamma^2$). **[OBSERVATION]** On every instance tested, the transcendence class is set by the number of two-center transition densities on distinct center pairs—zero or one gives genus zero, two gives genus one, while two densities on the same pair (the two-center exchange class) build the same fibre family yet close in the two-center seed set (Paper 18 [50]), so the fibre genus alone does not decide the class—exactly as for the repulsion integral.

[OBSERVATION] The consequence is a *unification*, not an escape. Explicitly-correlated integrals are native to the Fock momentum representation and cost the same as the corresponding repulsion integrals, because here, for the physical integrals tested, the curve of a two-electron integral is built by the densities, not by the operator between them; the kernel's mass can still move the class of the integrated object (the observation closing this section). In particular the elliptic obstruction of this paper—the genuine three-center, two-two-center-density integral—gates explicitly-correlated *evaluation* as well: the corresponding closed form (the kernel changes the kind of period, not the curve) that would evaluate the three-center repulsion integral would evaluate the multi-center F12 correction. What correlation buys is not a cheaper integral but a far more compact basis—a single r_{12} factor recovers the electron–electron cusp that orbital products approach only as $\ell \to \infty$—so the route trades many repulsion integrals for few correlated ones drawn from the same transcendence classes. We do not here treat the many-electron (resolution-of-identity) machinery of a full F12 method, nor a resource comparison with tuned Gaussian F12; the statement is about the two-electron integrals, whose momentum-space transcendence we have mapped.

[PRIOR ART + SYMBOLIC] The two-center integral with a Slater geminal shows where this reading needs care. Take the two-electron master integral of a diatomic molecule, with an inverse power and an exponential on each of the five distances. Without a geminal its equation in the bond length has second order and Bessel solutions [41]; with one it has fourth order, and Lesiuk and Moszynski defined its solutions by series and named them hyper-Bessel functions [40]. Their operator is Eq. (9) after a rescaling of the bond length. Both have the form $r\,\Delta(d/dr) + \frac{1}{2}\Delta'(d/dr)$ for an even quartic Δ, the Laplace image of a maximal cut with exponent $-1/2$. Here Δ is four times the Gram determinant of the loop momenta of the tetrahedron, with every line on shell; that tetrahedron is the Laplace transform of the master integral [40]. The geminal's exponent is the parameter: as it goes to zero, $\rho \to 0$ and the operator falls to the Bessel one, as $N \to K_0$ above. Both densities of the master lie on one pair of centers, so for this operator the kernel does change the class: the same-pair collapse is a property of the Coulomb kernel. Symmetric exponent patterns, the James–Coolidge and Kołos–Wolniewicz ones and the single-exponent $(AA|AA)$ and $(AB|AB)$ classes, give $\rho = 1$, the equal-scale collapse of Eq. (5); the $(AA|BB)$ and $(AA|AB)$ patterns and unequal exponents give a genus-one operator. Caps: this classifies the master's operator, not the physical integrals, which are its derivatives in the exponents; the master's source term was not re-derived, though the master's published values at its two test points were reproduced to 10^{-15} by an independent momentum-space route; no earlier statement of the identification was found in a short search (record `docs/predictions/2026-10-27_h2_geminal_elliptic.md`, registered before computing; guard `tests/test_paper59_h2_geminal_master.py`).

[OBSERVATION + PRIOR ART] Read as a Feynman graph, the cases of this section fit one pattern with the rungs beyond the third center, which Paper 63 [1] states as a bounded conjecture: the class is set by the shape of the density graph and the masses on its lines.

VIII. SCOPE, AND THE OPEN FRONTIER

Scope. The demonstration is for $1s$ orbitals at a common exponent. Higher (n, ℓ, m) multiply the integrand of Eq. (3) by solid-harmonic polynomials in $\boldsymbol{k}$ and shift the exponents; these change the weights and arguments but not the genus of the curve (5), which is set by the two distinct scales (Paper 63 [1], the dictionary). The result is a statement about the transcendence *class* of the radial core of the three-center integral, established on its simplest genuine instance; the integrated value, as a function of the angle, leaves the two-center class (below). It does not by itself deliver a polyatomic energy: a molecular Hamiltonian requires the full three-center tensor, and a value that is part closed-form and part numerical is neither more accurate (the Gaussian fits are already exact to fit quality) nor decidable. What the reduction buys is an accurate, fast, geometry-native evaluator for the three-center integral and—the point of this paper—a precise diagnosis of why it is hard.

The wall is closed-form, not computational. **[MEASURED]** The infinite series that the standard verdict falls back on is, concretely, the expansion of both densities in multipoles about the shared center (the Barnett–Coulson / Löwdin α-function route [8]; a recent analytic treatment of the same integrals, for linear geometries, is Ref. [7]; both are the numerical treatments named in Sec. VI). Each channel is a product of two one-pair factors, and the third center enters only through the relative orientation of the two pairs. On the water geometry ($R_{\mathrm{OH}} = 1.809$, $104.5°$) with O $1s/2s/2p$ and H $1s$ Slater orbitals, seven three-center integrals of both shared-O and shared-H type converge to $\sim 10^{-9}$ by $L \approx 12$. Gaussian-fit sequences converge onto every value; for the O $1s$ core the Gaussians approach it from below and are still 6×10^{-8} short at 18 Gaussians, while the channel sum is stable to thirteen digits (`tests/test_pairwise_general.py`,

tests/test_pairwise_three_center.py). **[MEASURED]** The channel sum is the Legendre series of the integral in $\cos\gamma$, γ the bond angle, so the decay of its terms locates the nearest singular point of the integral as a function of the angle. In every case tested the terms keep one sign, so that point is not the linear geometry. At water's equal bond lengths they fall like a power, about ℓ^{-8}, which puts it at $\gamma = 0$, where the two hydrogens would merge; at unequal lengths $D_1 \neq D_2$ they fall geometrically, at a rate within 3% of $D_</D_>$, the rate of $1/|Y-Z|$, whose only singular point is the two outer centers merging. Bent water is an ordinary point. Caps: the registered tolerance for that rate was 1% and it was missed in all four cases; $1s$ orbitals with two exponent sets only (record docs/predictions/2026-10-27_water_angle_singularities.md, registered before computing; guard tests/test_paper59_water_bond_angle.py). **[SYMBOLIC + MEASURED]** The singular term itself is derived. Only the two hydrogen cusps, meeting through $1/r_{12}$, make the integral non-analytic in the angle, and the lowest such term is $C\,|Y-Z|^7$ with $C = \zeta_{\mathrm{H}}^5\zeta_{\mathrm{X}}^3 e^{-\zeta_{\mathrm{X}}(D_1+D_2)}/630$. The same local computation, run on the two-center Coulomb integral of two $1s$ densities, reproduces the first two odd terms of its closed form, $(2/315)R^7 + (16/14175)R^9$, exactly. The term factors into one number per hydrogen. Call g_Y the coefficient of $|\mathbf{r}-Y|$ in the density $\chi_X\chi_Y$ at Y, its contact; Kato's cusp rule sets it to $-\zeta_{\mathrm{H}}\chi_X(Y)\chi_Y(Y)$. Then the term is $(\pi^2/630)\,g_Y g_Z\,|Y-Z|^7$, and $\pi^2/630$ is the same for any two densities with a point cusp. For Slater orbitals the π^2, a Fourier-measure factor, cancels against the normalizations, so C is π-free. This is the Coulomb form of Tan's contact relations, in which one short-distance number per cusp, times a universal function, fixes a slow tail [16]. The small-R series of two-center Coulomb integrals, with a non-analytic family of terms beside the even one, is Silverstone's [19]. For the one-electron analogue, the three-center nuclear-attraction integral, Bouferguene and Jones derived the large-ℓ terms of the same multipole series: geometric in the ratio of the two distances, times a power of ℓ set by the orbital, and only logarithmically convergent when the distances are equal [17]. The two-electron term was found first by Matuzas [18], whose preprint we found only after this work. For $(1s_A 1s_B|1s_A 1s_C)$ expanded about the shared center, the preprint proves a convergent germ in $|B-C|^2$ with the half-integer power 7/2 and no logarithms, the power 3/2 for the one-electron analogue, the same constant 1/630 at equal distances, and the decay laws ℓ^{-8} and ℓ^{-4} times the distance ratio below; it reads the tail, as a heuristic, from the cusp slopes on the two outer centers. Our earlier literature search (record docs/predictions/2026-10-28_water_landau_gaussian_fresh.md) missed it. The full text of Ref. [19] was not read, so whether it gives the two-center R^7 coefficient in this form is unchecked. The term fixes the power: ℓ^{-8} at equal lengths, and ℓ^{-4} times the geometric rate at unequal ones. At water's equal lengths the first two orders, $\alpha_7(1-x)^{7/2} + \alpha_9(1-x)^{9/2}$, both in closed form from the two exponents and the bond length, reproduce the coefficients to 10^{-4} at $\ell = 60$ with no fitted number, and what is left falls like ℓ^{-4}, the order of the first term left out. The rate test above missed because its fit left the power free: on a function with exactly this singularity that fit is 5% off, and with the derived power the four unequal-length rates come within 0.03% (reported only; that run was voided by one of its own controls; records docs/predictions/2026-10-27_water_cusp_singularity.md and docs/predictions/2026-10-27_water_cusp_next_term.md, each registered before computing). **[SYMBOLIC + MEASURED]** At unequal lengths the singular point moves to $x^* = (D_1^2 + D_2^2)/(2D_1D_2) > 1$, where $|Y-Z|^2 = 0$ at a complex angle, and every gradient of the two cusp functions enters at the same order there, so the endpoint value of C does not carry over. Summed, those orders are the moments of $24su$ on the simplex $s, u \geq 0$, $s+u \leq 1$: the term is $\Phi\,(2D_1D_2)^{7/2}(x^*-x)^{7/2}$ with $\Phi = (\pi^2/630)\,24\iint su\,G_Y(-sw)\,G_Z(uw)\,ds\,du$, $w = Y-Z$, the two cusp functions averaged along the segment between the hydrogens. This is the collinear pinch of the Landau equations for $|\mathbf{a}|$, $|\mathbf{b}|$ and $1/|w+\mathbf{a}-\mathbf{b}|$. For $1s$ functions Φ is a two-dimensional integral of $e^{-\zeta_{\mathrm{X}}(R_1(s)+R_2(u))}$ with $R_1(s)^2 = (1-s)D_1^2 + sD_2^2$, and it equals C at $D_1 = D_2$. Extracted from the coefficients with the subleading powers as fitted nuisances, the leading size matches Φ to 3.2×10^{-5} in four cases registered before computing, with both lengths and the hydrogen exponent varied and nothing fitted in the size; the endpoint value is off by 6.5 to 25% (record docs/predictions/2026-10-28_water_landau_gaussian_fresh.md, a first registration stopped on a control bound set without measuring, record docs/predictions/2026-10-28_water_landau_unequal.md). The pinch mechanism is standard, and taking the coefficient at one point is the rule for a simple pinch; here the pinch points fill a segment. The same size, as a triangle integral of the two oxygen exponentials, is in Ref. [18]; its reading as a Landau pinch was not found in a literature search (same record). Caps: $1s$ functions, oxygen exponents 3 and 4. **[SYMBOLIC + MEASURED]** With zero contact the ℓ^{-8} term is absent. For a hydrogen function whose first-order cusp cancels, the derived next term, $(1-x)^{9/2}$ with one such function and $(1-x)^{11/2}$ with two, reproduces the coefficients with nothing fitted, with the sign the derivation gives: to 0.5–1.9% at $\ell = 35$ on four cases registered before computing (two cusp-free pairs, two oxygen exponents, two bond lengths; record docs/predictions/2026-10-27_water_zero_contact_fresh.md). A first registration stopped on a sign clause it stated wrongly; its size and remainder clauses held (record docs/predictions/2026-10-27_water_zero_contact.md). Caps: symmetric geometry, $1s$ on the oxygen, cusp-free pairs of $1s$ functions only. **[SYMBOLIC + MEASURED]** Hydrogen p functions, the polarization functions of real water bases, have no $1s$-type contact,

so the reading predicts a different tail. A p_σ function on Y pointing along the bond is $\hat{\mathbf{y}} \cdot \nabla_Y f(|\mathbf{r} - Y|)$ with $f(s) = (s/\zeta + 1/\zeta^2)e^{-\zeta s}$, which is cusp-free, so every case is a derivative of the $1s$ integral in a bond length (generating higher angular momentum by differentiating in the nuclear coordinates is standard [13]), and the germ of each case follows from the cusp-free germs above. With one p_σ (the other hydrogen $1s$) or two, the coefficients keep $\rho^\ell \ell^{-4}$ at unequal lengths and fall like ℓ^{-10} at equal lengths, two powers faster than $1s$; with two out-of-plane p_π functions they fall like $\rho^\ell \ell^{-5}$ at unequal lengths. From exact coefficients to $\ell = 150$, extrapolated on $\ell = 100$–150, all five decay powers came within 0.002 of these values (bound 0.02), in two geometries registered before computing, with the $1s$ controls at -3.99965 and -8.00004 against -4 and -8 (record `docs/predictions/2026-10-29_hydrogen_p_functions.md`). Ref. [18] states higher angular momentum is outside its scope; the power rule used is its theorem, applied to the derived germs. Caps: $1s$ on the oxygen, one hydrogen exponent, p_π at equal lengths not scored, sizes not predicted. **[MEASURED]** Gaussian functions have no cusp, so the contact reading says their bond-angle series has no power-law tail. For one s Gaussian per center the integral is a Boys function, linear in $\cos\gamma$, and its coefficients fall like $T^\ell/(2\ell+1)!!$. STO-3G and STO-6G contractions follow the Slater ℓ^{-8} coefficients up to a crossover $\ell^* \approx 3.6$–$3.7\, D\sqrt{\alpha_{\max}}$, $\alpha_{\max}$ the tightest hydrogen exponent, and fall away past it. A model that keeps only the cusp-cusp term placed ℓ^* to within one in four cases registered before computing, and ℓ^* moves with $\sqrt{\alpha_{\max}}$ (record `docs/predictions/2026-10-28_water_landau_gaussian_fresh.md`; a first registration stopped on a control whose precision it did not set, record `docs/predictions/2026-10-28_water_gaussian_contact.md`). That a Gaussian fit copies a cusp only down to about $1/\sqrt{\alpha_{\max}}$ is basis-set lore; this is a check of the contact reading, not a new result. Caps: symmetric geometry, s functions, two published contractions. **[SYMBOLIC + MEASURED]** The bond-angle function obeys a linear differential equation in x, since the integrand is holonomic in the positions; we looked for it. Each Legendre coefficient is a finite sum of rational numbers times fixed constants, built from the three-term rules of i_ℓ and k_ℓ in ℓ and exact integration by parts: six constants for the one-electron three-center integral $\int \phi_X \phi_Y / |\mathbf{r} - Z|$ (four exponentials and $e^{\mp\zeta_H D_1} E_1((\zeta_X + \zeta_H)D_2)$), and 31 for $(XY|XZ)$, all exponentials times 1, E_1 or a logarithm, so each channel stays in the two-center class. Silverstone wrote a related three-center $1s$ integral, $(1s_a 1s_a | 1s_b 1s_c)$, as such a Legendre series in the angle between the bond vectors, every term in closed form [12]; the fixed constant set, independent of ℓ, is what is new here. With the coefficients exact, the equation is found by linear algebra modulo primes, where an approximate fit cannot imitate it; from floating-point coefficients at 110 to 200 digits, approximate low-order fits matched even a known control to the data's precision and hid it. The one-electron integral obeys an operator of order 8 with polynomial coefficients of degree 31, the same in two geometries and modulo two primes; an independent derivation from the ℓ-recurrences alone, with no fit to the coefficients, gives an order-9 operator that the order-8 one divides exactly. Its only finite singular point with a non-integer exponent is x^*, with exponents $3/2$ and $5/2$ besides integers: Poisson's equation turns the density's $|\mathbf{r} - Y|$ cusp term into a $|Z - Y|^3$ term of the potential. The points $x = \pm 1$ carry repeated integer exponents and one further rational point an exponent -2; the integral is analytic at both, so these belong to other solutions. Infinity is irregular. For $(XY|XZ)$ no operator exists with fewer than about 1970 unknowns at any order up to 40 (2001 coefficients, two geometries, two primes); the constant components alone carry seven distinct operators of orders 6 to 13 (records `docs/predictions/2026-10-28_nuclear_attraction_angle_opera`, `docs/predictions/2026-10-28_bottom_up_angle_operator.md`, `docs/predictions/2026-10-28_two_electron_angle_operator_ex` and `docs/predictions/2026-10-28_angle_operator.md`, each registered before computing). Built up from the ℓ-recurrences in the same way, with no fit, the $(XY|XZ)$ coefficients obey a recurrence in ℓ of order 41 and degree 926 (one geometry, two primes; it annihilates the exact coefficients up to $\ell = 400$). Among its rational characteristic roots, only those at x^* carry exponents of a branch point. Evaluated to $\ell = 300$ with more than 550 digits, the exact coefficients decay as $\rho^\ell \ell^{-4}$, $\rho = D_</D_>$, so the slowest bond-angle term is $(x^* - x)^{7/2}$, the cusp-cusp term $|Y - Z|^7$ above. The recurrence's other solutions, among them ℓ^3 and ℓ^1 at x^*, carry zero weight (records `docs/predictions/2026-10-28_two_electron_bottom_up.md` and `docs/predictions/2026-10-29_two_electron_large_l.md`, both registered before computing). Guessing an operator from many exact terms is standard [37, 38]; an equation in the bond angle for a three-center integral was not found in a literature search. Caps: $1s$ functions; the constants are assumed linearly independent over $\mathbb{Q}$. Since every channel stays in the two-center class, any higher-class content of the integrated value must come from the infinite channel sum. **[MEASURED]** It does, as a function of the angle. A two-center-class form in $w = |Y - Z|$ is a finite sum of rational functions, exponentials $e^{\lambda w}$ and their iterated integrals, so its operator in w factors completely into first-order pieces. For each of the 31 constant components we built the operator from the exact coefficients modulo a prime and removed first-order right factors one at a time. At both geometries and modulo both primes, six components stop, after none or two such steps, at a piece of order 6 with no first-order right factor. Under the assumption that the 31 constants are linearly independent over $\overline{\mathbb{Q}}$, the integrated $(XY|XZ)$, as a function of the angle, therefore leaves the two-center class, and with it the class of iterated integrals of exponentials and rational functions. The one-electron integral leaves it too: at

one geometry all six of its components stop with no step taken, modulo both primes, and at the other geometry the test is undecided. We had predicted the opposite for the one-electron integral. This is a statement about the function of the angle: the class of the value at a single geometry is not decided, and whether the blocking piece is elliptic is a reading of local data, not a proof. The search for first-order factors is incomplete where a singular factor does not split modulo the prime, since a first-order factor may need an algebraic extension of the constants [39]. Unsplit exponential rates are flagged and such components not counted; a planted control shows that unsplit singular points can also hide a factor without a flag, and none of the counted components has such a point with a fractional exponent. Caps: $1s$ functions, two geometries, two primes, minimal operators not certified (record `docs/predictions/2026-11-01_integrated_class.md`, registered before computing). The radial core is a period of an irreducible connection (Sec. V, **[SYMBOLIC + MEASURED]**), so it has no closed form through any factorization; that does not bar an accurate evaluation, and whether the integrated three-center integral has a finite closed form is the open problem below.

The open problem. As shown in Sec. V, the elliptic-dilogarithm route is obstructed: $N(D)$ is not an elliptic dilogarithm of elementary or period-level data but a period of the rank-four irregular connection (9). We settle the differential-Galois *reducibility*: the fourth-order operator is irreducible over $\mathbb{C}(D)$—hence over $\mathbb{Q}(\rho)(D)$—because it is the Fourier–Laplace transform of an irreducible connection, confirmed concretely by its exact-integer monodromy (Sec. V), so $N(D)$ admits no closed form through any factorization of the operator. The one remaining question is whether the irreducible connection nonetheless carries a *transcendental* closed form, which we do not settle. As a function of the bond angle the integrated object is large: its operator in x has more than about 1970 unknowns at every order up to 40, where the one-electron analogue needs order 8, though its Legendre coefficients obey a recurrence in ℓ of order 41 derived from the ℓ-recurrences (Sec. VIII). The companion (Paper 61) localizes it: the object is an Eisenstein / complex-multiplication Bessel-moment period—not a cusp-form L-value and not an elliptic dilogarithm—whose master family obeys the quadratic period relations $B = \pi\,\Omega$, Ω the canonical symplectic intersection form now identified in closed form (developed in the companion, Paper 61), preserved by the integral monodromy; the open piece is the explicit finite reduction of the observable itself to π and the CM-fibre Γ-values. (This tempers a literature-scout expectation that the object would sit at the tractable elliptic-dilogarithm level because it has only two propagator factors; that pattern-match from the sunrise is overridden by the in-module computation of Sec. V.) Because it continues the Averys' momentum-space Sturmian program and requires elliptic-Feynman-integral technique one level beyond the sunrise, it is the natural place for collaboration.

IX. CONCLUSION

The radial core of the three-center electron-repulsion integral is not merely a harder version of the two-center one; it is an object of higher genus. Two centers close at genus zero over $\{E_1, \ln, \gamma\}$; three centers, read in momentum space where the coordinate obstruction dissolves, reduce to a two-scale Bessel moment on an elliptic curve, an elliptic deformation of the Bessel K_0 with an explicit fourth-order Picard–Fuchs equation and a period that is a complete elliptic integral. At the level of the radial core, the seventy-year difficulty of polyatomic Slater integrals is a genus jump; as a function of the bond angle the integrated value leaves the two-center class too (Sec. VIII); and a classical molecular integral turns out to be an instance of the modern theory of elliptic Feynman integrals—a bridge that, to our knowledge, had not been drawn. With a Slater geminal the two-center master integral already reaches the same curve: its published fourth-order operator is the three-center one rescaled, and only symmetric exponent patterns collapse. The finite closed form remains open—and, we have shown, past the reach of the sunrise's elliptic-dilogarithm mechanism, whose inhomogeneity is absent here: the object is a period of a rank-four irregular connection which we prove irreducible—it is the Fourier–Laplace transform of an irreducible connection, and its monodromy admits no invariant coordinate subspace—so it does not close by any factorization. Whether it closes at the irreducible level—as an Eisenstein/CM Bessel-moment period, a finite reduction to π and the CM-fibre Γ-values (companion, Paper 61)—is the one frontier that remains, posed here as it deserves to be. What lies beyond the third center—how the rung climbs with the number of densities an integral links, and how far the reading reaches past $1s$ orbitals—is the subject of Paper 63 [1].

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